

Week 8

MATH503

Hypothesis testing

Housekeeping!

- Keep up with the Myimaths tasks (25% of final grade)
- Attend the tutorials (30% of final grade)
- Check the grades on Canvas for the group activities
- Start with an exam summary of formulas and key examples
- Get to know the manual

Hypothesis testing

Hypothesis testing is the second most common method of statistical inference used to draw statistical conclusions.

It allows us decide in **favour or against** a particular claim.

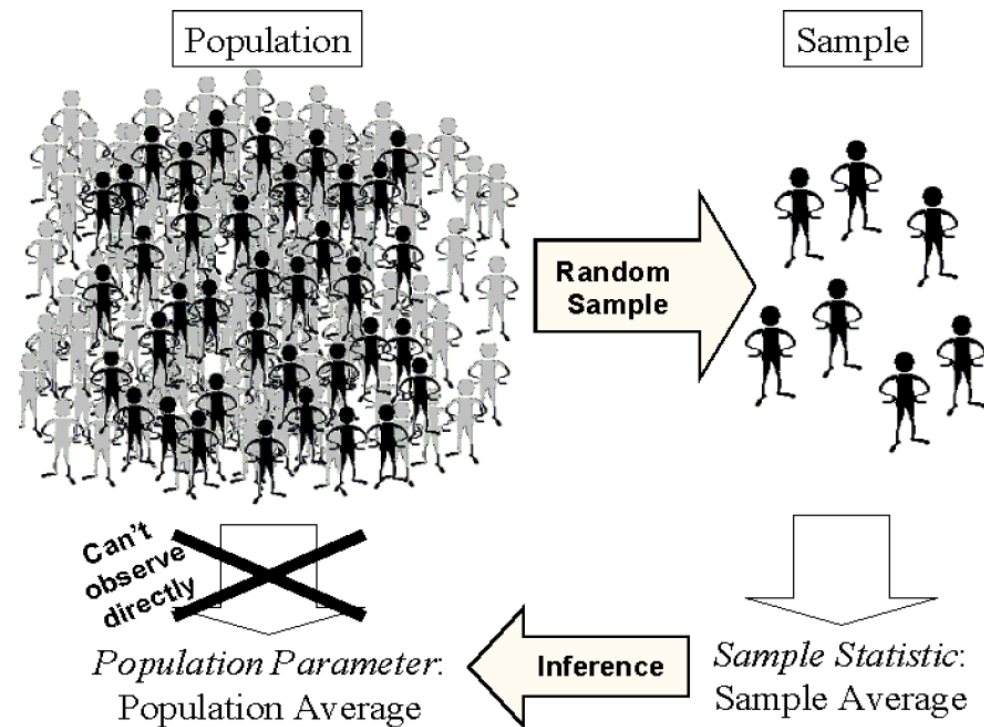
The claims are opposing statements about a particular feature of the population under study.

In MATH503 we will use hypothesis testing for

- the population proportion (p)
- the population mean (μ).

Populations and samples

Hypothesis-testing procedures rely on using the information in a **random sample from the population of interest.**



Null hypothesis & Alternative hypothesis

Null Hypothesis:

The claim that we weigh evidence against is called the **null hypothesis (symbol H_0)**.

Null hypothesis: Historical value, claim or product specification.

Alternative Hypothesis:

The claim that we are trying to find evidence for, what you want 'to prove', is the **alternative hypothesis (symbol H_a or H_1 or H_A)**.

Stating Hypotheses – Example 1

A free-throw shooter claims that his long-run proportion of made free throws is $p = 0.8$, but we suspect that the player is exaggerating.

Let 'p' denote the shooter's long-run proportion of made free throws.

In words:

H_0 : *(known facts) Shooter's long-run proportion is 0.8*

H_a : *Shooter is exaggerating ($p < 0.8$)*

In symbols:

$H_0: p = 0.80$

$H_a: p < 0.80$

(Note: There is no info yet to do hypothesis testing.)

Stating Hypotheses – more options

There are three options for H_a

Option 1: The shooter is exaggerating his shooting ability

H_0 : (known facts) Shooter's long-run proportion is 0.8

H_a : Shooter is exaggerating his ability ($p < 0.8$)

In symbols:

$H_0: p = 0.80$

$H_a: p < 0.80$

Option 2: The shooter is understating his shooting ability

H_0 : (known facts) Shooter's long-run proportion is 0.8

H_a : Shooter is understating his ability ($p > 0.8$)

In symbols:

$H_0: p = 0.80$

$H_a: p > 0.80$

Option 3: The shooter long-run proportion is no longer 0.8.

H_0 : (known facts) Shooter's long-run proportion is 0.8

H_a : Shooter's long-run proportion has changed ($p \neq 0.8$)

In symbols:

$H_0: p = 0.80$

$H_a: p \neq 0.80$

FACT: You have to decide. Read the story and make a decision about H_a .

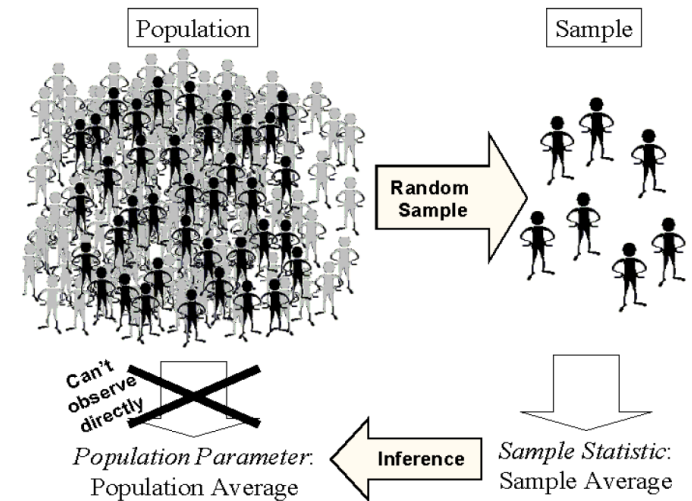
This is part of the hypothesis test.

Stating Hypotheses – Example 2

After studying a sample of 150 students at AUT-ECMS, a researcher concludes that students study an average of 10.25 hours per week during the academic year. It is known, from research conducted in 2018 – 2019 that AUT students, study on average 8.75 hrs/week.

Identify

- the population
- the sample
- the sample statistic
- the population parameter



Stating Hypotheses – Example 2

Let μ be the average number of hours/week that AUT students spend studying

Option 1:

H_0 : (known fact) AUT students study on average 8.75hr/week

H_a : AUT students study on average > 8.75 hr/week

$H_0: \mu = 8.75$

$H_a: \mu > 8.75$

Option 2:

H_0 : (known fact) AUT students study on average 8.75hr/week

H_a : AUT students study on average < 8.75 hr/week

$H_0: \mu = 8.75$

$H_a: \mu < 8.75$

Option 3:

H_0 : (known fact) AUT students study on average 8.75hr/week

H_a : The average hr/week that AUT students study is no longer 8.75hr/week

$H_0: \mu = 8.75$

$H_a: \mu \neq 8.75$

One-sided and Two-sided Hypothesis tests

The alternative hypothesis is **one-sided**

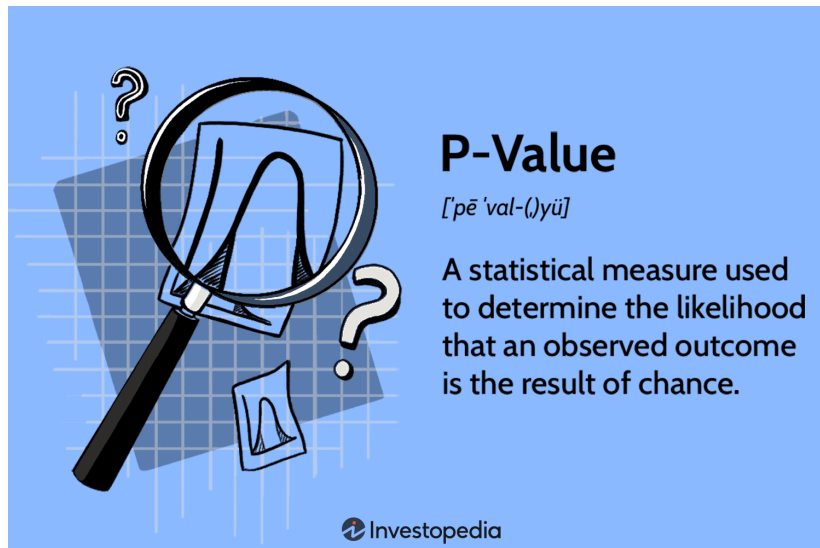
- if it states that a parameter is *greater than* the null value
- if it states that the parameter is *less than* the null value
- Option 1 and Option 2 in the examples

The alternative hypothesis is **two-sided**

- if it states that the parameter is *different from* the null value
- Option 3 in the examples

Definition of 'p-value'

The 'p-value' is the **probability** of obtaining **test results at least as extreme as the result actually observed**, **given that the null hypothesis is true**.



For alternative hypothesis $H_a: p < \dots$ or $\mu < \dots$, the more extreme cases are in the left tail.

For alternative hypothesis $H_a: p > \dots$ or $\mu > \dots$, the more extreme cases are in the right tail.

For alternative hypothesis $H_a: p \neq \dots$ or $\mu \neq \dots$, the more extreme cases are in the both tails.

Significance level of a hypothesis test

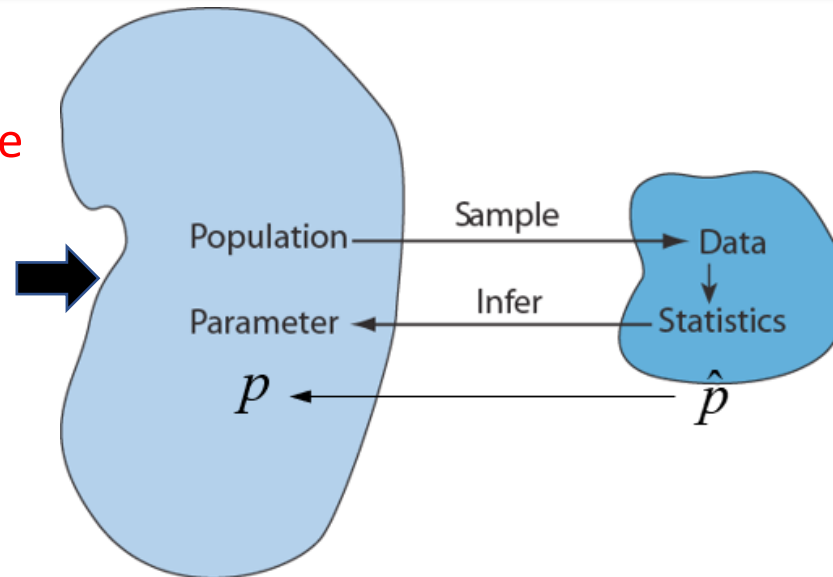
The threshold for statistical significance is arbitrary.

The most common threshold is 0.05, which means that the data is likely to occur less than 5% of the time under the null hypothesis.

An election candidate believes she has the support of 40% of the residents in a particular town. A researcher wants to test, at the 5% significance level, whether the candidate is over-estimating her support. The researcher asks 20 people whether they support the candidate or not. Three people say they do.

- a) Is this problem about a proportion or a population mean?
- b) Write down a sensible null hypothesis along with an alternative.
- c) Explain the condition under which the null hypothesis would be rejected.
- d) Write down your conclusions based on the p-value.
- e) What is the critical region?

The 40% refers to the whole population of the town



The 20 people are a sample from the town.

We will now make inference about the whole town based on this sample.

Null hypothesis & Alternative hypothesis

Null Hypothesis:

The claim that we weigh evidence against is called the **null hypothesis** (symbol H_0).

Null hypothesis: Historical value, claim or product specification.

Alternative Hypothesis:

The claim that we are trying to find evidence for, what you want 'to prove', is the **alternative hypothesis** (symbol H_a or H_1 or H_A).

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- a) Is this problem about a proportion or a population mean?
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- d) Write down your conclusions based on the p-value.
- e) What is the critical region?

a) Population proportion

b) Let p = 'proportion of supporters'

In words:

H_0 : 40% of the residents in the town support the candidate

H_a : Less than 40% of the residents in the town support the candidate

In symbols

$H_0: p = 0.4$

$H_a: p < 0.4$

Definition of 'p-value'

The 'p-value' is the probability of obtaining **test results at least as extreme as the result actually observed**, given that the null hypothesis is true.

Significance level of a hypothesis test

Usually, hypothesis testing is done with a 5% significance level

Decision about the null hypothesis

If $p\text{-value} < 5\%$, we reject the null hypothesis

If $p\text{-value} > 5\%$, we cannot reject the null hypothesis

An election candidate believes she has the support of 40% of the residents in a particular town. A researcher wants to test, at the 5% significance level, whether the candidate is over-estimating her support. The researcher asks 20 people whether they support the candidate or not. Three people say they do.

- c) Explain the condition under which the null hypothesis would be rejected.
- d) Write down your conclusions based on the p-value.
- e) What is the critical region?

c) In the sample 3 out of 20 people support the candidate.

The p-value, is the probability of obtaining three supporters or fewer, given that 40% of the residents support her ($p = 0.40$).

p-value = $P(\text{three supporters or less out of 20} \mid \text{the proportion of supporters is } p = 0.4)$

We will reject the null hypothesis if the p-value < 0.05 .

An election candidate believes she has the support of 40% of the residents in a particular town. A researcher wants to test, at the 5% significance level, whether the candidate is over-estimating her support. The researcher asks 20 people whether they support the candidate or not. Three people say they do.

- d) Write down your conclusions based on the p-value.
- e) What is the critical region?

d)

p-value = $P(\text{three supporters or less out of 20} \mid \text{the proportion of supporters is } p = 0.4)$

Use the binomial distribution with $n = 20$ and $p = 0.4$:

p-value = $P(0 \text{ successes}) + P(1 \text{ success}) + P(2 \text{ successes}) + P(3 \text{ successes}) = 0.016$
 $\text{dbinom}(0,20,0.4) + \text{dbinom}(1,20,0.4) + \text{dbinom}(2,20,0.4) + \text{dbinom}(3,20,0.4)$ or $\text{pbinom}(3,20,0.4)$
p-value < 0.05 (significance level)

Reject the null hypothesis.

We conclude that the candidate's statement about the 40% support is incorrect.

She is over-estimating her support.

An election candidate believes she has the support of 40% of the residents in a particular town. A researcher wants to test, at the 5% significance level, whether the candidate is over-estimating her support. The researcher asks 20 people whether they support the candidate or not. Three people say they do.

e) What is the critical region? (Also called the rejection region.)

e) Critical region:

Find the largest number of successes out of 20 trials so that the cumulative probability remains less than 5%.

$$P(\text{at most } x \text{ successes out of } 20) < 0.05$$

$$P(x \leq 3) = 0.016 = 1.6\% < 5\% \text{ Reject null hypothesis } \text{pbinom}(3,20,0.4)=1.6\%$$

$$P(x \leq 4) = 0.051 = 5.1\% > 5\% \text{ Cannot reject the null hypothesis } \text{pbinom}(4,20,0.4)=5.1\%$$

The critical region

If 3 or fewer people (out of 20) support the candidate, we reject the null hypothesis.

Hypothesis testing for population mean

Marks in an exam are normally distributed with $\mu = 56.9$ and $\sigma = 6.9$.

A teacher looks at the marks of 11 of her pupils. Their mean mark is 56.

She decides to test if the mean is lower than 56.9 at a 5% significance level.

Myimaths: Hypothesis testing 3

Null hypothesis:

H_0 : The mean mark for the test is 56.9

$$H_0: \mu = 56.9$$

Alternative hypothesis:

H_a : The mean mark for the test is less than 56.9

$$H_a: \mu < 56.9$$

Hypothesis testing for population mean

Marks in an exam are normally distributed with $\mu = 56.9$ and $\sigma = 6.9$.

A teacher looks at the marks of 11 of her pupils. Their mean mark is 56.

She decides to test if the mean is lower than 56.9 at a 5% significance level.

$$H_0: \mu = 56.9 \quad \text{and} \quad H_a: \mu < 56.9$$

One tail test (left tail) at 5% significance level

Test statistic: $\bar{x}$ has a normal distribution with mean 56.9 and $sd = \frac{\sigma}{\sqrt{n}} = \frac{6.9}{\sqrt{11}} = 2.08$

Three methods:

1. Calculate the p-value and compare with the significance level (preferred method)
2. Find the rejection region and check if $\bar{x}$ is in the rejection region
3. Compare the Z-score for $\bar{x}$ with the critical z-score value

Hypothesis testing for μ : p-value

Marks in an exam are normally distributed with $\mu = 56.9$ and $\sigma = 6.9$.

A teacher looks at the marks of 11 of her pupils. Their mean mark is 56.

She decides to test if the mean is lower than 56.9 at a 5% significance level.

$$H_0: \mu = 56.9 \quad \text{and} \quad H_a: \mu < 56.9$$

One tail test (left tail) at 5% significance level

Test statistic: $\bar{x}$ has a normal distribution with mean 56.9 and $\text{sd} = \frac{6.9}{\sqrt{11}} = 2.08$

p-value = $P(\bar{x} < 56) = 0.3326 > 0.05$ (significance level) $\text{pnorm}(56, 56.9, 2.08) = 0.33$

No evidence to reject the null hypothesis
(no evidence that the mean is lower than 56.9)

Hypothesis testing for μ : Rejection region

Marks in an exam are normally distributed with $\mu = 56.9$ and $\sigma = 6.9$.

A teacher looks at the marks of 11 of her pupils. Their mean mark is 56.

She decides to test if the mean is lower than 56.9 at a 5% significance level.

$$H_0: \mu = 56.9 \quad \text{and} \quad H_a: \mu < 56.9$$

One tail test (left tail) at 5% significance level

Test statistic: $\bar{x}$ has a normal distribution with mean 56.9 and $\text{sd} = \frac{6.9}{\sqrt{11}} = 2.08$

Critical region: $P(\bar{x} < \text{critical value}) = 0.05$

Use inverse normal distribution: **critical value = $\text{qnorm}(0.05, 56.9, 2.08) = 53.4787$**

Rejection region is $\bar{x} < 53.4787$

No evidence to reject the null hypothesis

(no evidence that the mean is lower than 56.9)

Hypothesis testing for μ : Z score

Marks in an exam are normally distributed with $\mu = 56.9$ and $\sigma = 6.9$.

A teacher looks at the marks of 11 of her pupils. Their mean mark is 56.

She decides to test if the mean is lower than 56.9 at a 5% significance level.

$$H_0: \mu = 56.9 \quad \text{and} \quad H_a: \mu < 56.9$$

One tail test (left tail) at 5% significance level

For $Z \sim N(0,1)$ the critical Z- value is -1.6449 as $P(Z < -1.6449) < 0.05$

Test statistic: $\bar{x}$ has a normal distribution with mean 56.9 and $sd = \frac{6.9}{\sqrt{11}} = 2.08$

$$\text{Find the Z-score: } Z = \frac{\bar{x} - \mu}{sd} = \frac{56 - 56.9}{2.08} = -0.4327 > -1.6449$$

No evidence to reject the null hypothesis
(no evidence that the mean is lower than 56.9)

Example: weights of kiwi birds (in manual)

An ecologist catches kiwi birds in a particular forest and weighs them. The data in kg is as follows:

1.08, 1, 0.96, 0.84, 0.91, 1.21, 1.03, 1.08, 1.09, 1.04, 0.95, 0.87, 1.09, 1.09, 1.08, 0.93, 1.07, 1.16, 1.21, 0.94, 1.16, 0.92, 1.09, 1.01, 1.08, 1.07, 0.89, 1.02, 1.14, 0.9

Nationally, the weight of a kiwi bird is known to have a mean of 1kg.

Question: Is there any evidence for the average weight of the kiwi in this forest to differ from the national mean of 1kg?

Example: weights of kiwi birds

- a) Is this problem about a proportion or a population mean?
- b) Write down a sensible null hypothesis along with an alternative.
- c) Explain the condition under which the null hypothesis would be rejected.
- d) Write down your conclusions based on the p-value.

Answers

a) About a population mean

Population?

Sample?

Observation?

Example: weights of kiwi birds

- a) Is this problem about a proportion or a population mean?
- b) Write down a sensible null hypothesis along with an alternative.
- c) Explain the condition under which the null hypothesis would be rejected.
- d) Write down your conclusions based on the p-value.

Answers

b) Let μ = 'mean weight of kiwi in this forest'

In words:

H_0 : the mean weight for kiwis in this forest is 1 kg

H_a : the mean weight for kiwis in this forest is different from 1 kg

In symbols

$$H_0: \mu = 1$$

$$H_a: \mu \neq 1$$

Example: weights of kiwi birds

- c) Explain the condition under which the null hypothesis would be rejected.
- d) Write down your conclusions based on the p-value.

Answers

c) In symbols: $H_0: \mu = 1$ and $H_a: \mu \neq 1$

First find the mean weight for the kiwi in the sample (use Rstudio):

$$n = 30 \text{ and } \bar{x} = 1.030333 \text{ and } sd = 0.1006696$$

For the sample mean: $\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

We will reject the null hypothesis if the p-value $< 5\% = 0.05$

Two sided-test

Example: weights of kiwi birds

d) Write down your conclusions based on the p-value.

Answers

d) In symbols: $H_0: \mu = 1$ and $H_a: \mu \neq 1$

$$\text{p-value} = 2 \times P(\bar{x} > 1.030333) \text{ with } \mu = 1 \text{ and } sd = \frac{0.1006696}{\sqrt{30}}$$

$$\text{p-value} = 0.099 > 0.05 \text{ (significance level)}$$

There is no evidence to reject the null hypothesis

There is no evidence that the average weight of the kiwi in this forest differ from the national average of 1kg.



What did you get out of today?

- What was the purpose of today's lecture?
- What is one key thing I learnt?
- Is there anything that puzzled me?